

Data Source Gatekeeper & Reference Microservice Development

BASIS SENTIMENT SERVICE · MAY 18 SPRINT

Sprint: May 18 – June 1, 2026 · Demo Day: June 12, 2026 · Issued: April 17, 2026

1. Work Order Overview

FIELD	DETAIL
Work Order	Data Source Gatekeeper & Reference Microservice Development
Sprint	BASIS SENTIMENT SERVICE — May 18, 2026 Sprint
Lead Tech	Dimitris (architecture direction + microservice spec sign-off)
Collaborators	Zilin Mao (commodity SME + COT scoring) · Jocelyn/Aras consult
Sprint Window	May 18 – June 1, 2026 (2 weeks)
Demo Day	June 12, 2026 — live BASIS SENTIMENT demo to Jim Marzano
Total Hours	~82 hours (2 weeks, estimated 41 hrs/week)
Primary Output	Production-ready data microservices for Tier 1 sources + curated catalog.json + VAULT integration endpoint

2. Sprint Introduction — Session 1 (1 Hour Required)

This section defines the required 1-hour onboarding and orientation session to be completed on Day 1 of the sprint (May 18). This session is not optional — it establishes the architectural frame, TOS compliance rules, and role expectations before any code is written.

2.1 Session Agenda (60 Minutes)

TIME	TOPIC	CONTENT
0:00–0:10	Jim: Commercial Context	Why BASIS SENTIMENT SERVICE matters commercially. Energy vertical TAM. FCM partner opportunity (StoneX, RJO). How a commodity trader experiences basis today (Mao narrates one real example from his Zhongcai Group trading history — corn or energy). What Metrixx adds that no current platform provides: AI-synthesized physical + options + COT sentiment in one subscriber output.
0:10–0:20	Jim: Product Architecture	Metrixx stack overview: Claude API → Gamma Playbook → VAULT → MY DESK. How the data microservices Park is building feed the Claude sentiment engine. How Mao's COT scoring rubric becomes a structured JSON payload that Claude uses to generate the 50-word basis alert. The jet fuel Europe shortage (April 2026) as a live prototype — reconstruct the signal from EIA + Reuters + ICE data.
0:20–0:30	Dimitris: Microservice Architecture Brief	Dimitris presents the microservice architecture spec: individual containerized services per data source (EIA, USDA, CFTC), unified JSON schema output, internal REST API contract, cron schedule, VAULT write endpoint. Park confirms understanding of each service boundary. No monolithic scripts — each source is an independent, deployable service. Discuss: error handling, retry logic, rate limit management for free-tier APIs.
0:40–0:50	All: Role Assignments + Q&A	Zillin: confirm Week 1 and Week 2 deliverables verbatim, confirm COT scoring rubric and energy or grain basis case study deliverables. no later than May 20 (Wednesday of Week 1). Jim: confirms demo format — live data, live Claude output, no static screenshots.

0:50– 1:00	Production Mindset Close	Jim closes with one sentence: "What you deliver on June 1 goes in front of FCM clients. Not a demo — a product." Emphasize that Metrixx has been in Claude API production for 14+ months. The expectation is production-grade output, not academic proof of concept. Park owns the data integrity of the entire service — every faulty data pull is a faulty subscriber output.
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3. Product & Platform Overview

This section provides the platform context Park needs to build microservices that integrate cleanly with the existing Metrixx production stack. Read before writing code.

3.1 What we’re building (and Why)

The BASIS SENTIMENT SERVICE delivers a Claude-generated intelligence output — a 50-character headline and a 50-word narrative — synthesizing physical commodity market data with Gamma Playbook options sentiment and COT positioning data. Park’s microservices are the data acquisition layer: they pull, normalize, and deliver structured JSON from Tier 1 government sources into the sentiment engine.

Park is not building the sentiment scoring logic (that is Wei + Mao). Park is building the pipes that make those outputs possible — reliably, on schedule, at production quality.

3.2 Platform Stack (Reference Only — Liu owns DevOps)

COMPONENT	DETAIL
AI Engine	Claude API (claude-sonnet-4-6) — Park’s JSON output is a payload element in the Claude system prompt for sentiment generation
Database	PostgreSQL (managed, DigitalOcean) — Park’s microservices write normalized data to tables defined in Dimitris spec
VAULT	Persistent intelligence hub — Park microservices write to VAULT endpoints; users recall BASIS reports by instrument/date
MY DESK	Rocket.Chat + React UI — BASIS SENTIMENT output surfaces in MY DESK BASIS panel after VAULT write
Hosting	Coolify/Linux (Liu manages) — Park microservices deployed as Docker containers per Dimitris architecture
Gamma Playbook	Park microservices must pass physical price data to Gamma Playbook’s existing JSON schema — confirm field names with Wei Week 1

4. Task Detail — Hours & Deliverables

4.1 Week 1: May 18 – 23 — Foundation Microservices

TASK	HRS	NOTES / ACCEPTANCE CRITERIA
Day 1 — Sprint intro session (Section 2)	1.0	Attend full 60-min session. Confirm role. No code before session complete.
Dimitris architecture meeting — microservice spec review	1.0	Receive: service boundaries, JSON output schema, PostgreSQL table spec, cron schedule, VAULT write contract. Document in service_spec_v1.md before coding.
EIA microservice — build & test	4.0	Python containerized service: gather ANY of WTI, Brent, Henry Hub, RBOB, Heating Oil, Jet Fuel series daily physical; use USDA for overview. Output: normalized JSON per Dimitris schema. Rate limit handling. Unit tests included. Mao validates price accuracy on Day 3.
USDA AMS grain elevator microservice	4.0	MARS API: corn elevator bids (Decatur/Gulf), soybean meal, HRW/SRW wheat. Output: normalized JSON with delivery point location tags. Mao validates grain basis calculation Day 3.
CFTC COT automated weekly pull	3.0	Disaggregated COT + Legacy COT weekly pull from data.cftc.gov. Cron trigger: Friday 4:00pm ET. Output: structured JSON by commodity (CL, NG, ZC, ZS, ZW, GC, SI) with Producer/Merchant and Managed Money net positions. Mao receives JSON to build COT scoring rubric.
Catalog.json schema — initial draft with Dimitris	1.5	Define catalog.json structure: source_id, tier, label, api_endpoint, update_freq, tos_status, last_reviewed_date, gatekeeper_sign_off. Submit to Dimitris for review by Friday May 22.
TOS compliance review — all Tier 1 and Tier 2 sources	2.0	Review TOS for each source in the Master Reference Catalog. Document clearance decision (GO / HOLD / REVIEW / RED). Confirm Argus and Platts are explicitly blocked. Submit to Jim for sign-off.
WEEK 1 TOTAL	16.5	Checkpoint: Jim reviews EIA and USDA microservice output JSON on Friday May 23. GO/NO-GO for Week 2.

4.2 Week 2: May 25 – June 1 — Integration & Catalog Completion

TASK	HRS	NOTES / ACCEPTANCE CRITERIA
Baker Hughes rig count — scheduled weekly pull	2.0	CSV download automation (Friday after close). Parse oil vs. gas rig count. Delta calculation week-over-week. Write to PostgreSQL. Flag -10 or more rigs as energy scarcity trigger in output JSON.
FRED macro overlay microservice	2.0	FRED API: DFF (Fed Funds), T10Y2Y, CPIAUCSL, PPIACO. Monthly and daily pulls per series. Macro overlay JSON delivered alongside basis data for MVIXX and interest rate basis context.
CME public settlement pull	2.0	Daily CME/NYMEX/CBOT settlement prices for front-month CL, NG, ZC, ZS, ZW, GC, SI, HO, RB.

		Automated end-of-day pull. Output: JSON with settlement price, volume, OI for each contract. This is the futures leg of the BASIS calculation.
Reuters/EIA narrative sentiment pipeline	3.0	RSS feed ingestion: Reuters Commodities + EIA Today in Energy. Parse headlines + snippets. Feed into Claude sentiment pre-processing layer. Mao reviews prompt structure for commodity context accuracy. Output: tagged event alerts with commodity, direction, and confidence score.
Jet fuel shortage cron case study — integration test	2.0	End-to-end test using April 2026 Europe jet fuel shortage as prototype. Pull EIA jet fuel series + ICE Gasoil settlement + Reuters RSS archive. Run through basis calculation + Claude sentiment generation. Produce sample 50-char / 50-word output. Mao validates output for market accuracy.
Catalog.json — final version + gatekeeper sign-off	2.0	Complete catalog.json with all 30+ sources TOS-reviewed and classified. Park signs as gatekeeper. Jim countersigns. Catalog committed to Metrixx repo. No production deployment of any source without catalog entry.
VAULT write integration — basis data archival	3.0	VAULT endpoint integration: each completed basis calculation (physical price + futures settlement + COT overlay + sentiment score) written to VAULT with instrument tag, delivery point, date, and source provenance. Liu reviews deployment. Users can recall any prior basis report from MY DESK.
WEEK 2 TOTAL	16.0	Checkpoint: full end-to-end demo ready June 1. VAULT populated. Claude generating basis alerts from live microservice data.

TOTAL SPRINT HOURS — WO-PARK-01

32.5

hrs / 2 wks

5. Microservice Architecture — Per Dimitris Direction

Architecture decisions belong to Dimitris. The following defines Park's implementation constraints. Park does NOT make architecture decisions unilaterally — all service boundary and schema decisions require Dimitris confirmation before coding begins.

5.1 Service Architecture Principles

- One containerized service per data source — EIA, USDA AMS, CFTC, Baker Hughes, FRED, CME settlements, and narrative feed are SEPARATE services, not a monolith
- Each service outputs a standardized JSON envelope: { source_id, instrument, delivery_point, timestamp, data_type, value, unit, tos_status, gatekeeper_cleared, raw_source_url }
- Services write to PostgreSQL via shared data access layer — schema defined by Dimitris, not Park
- Cron schedules per service — not all sources update daily; schedule must match source update frequency (see catalog)
- Error handling: exponential backoff on API failures, dead letter queue for failed pulls, alert to Jim/Liu on repeated failure
- Rate limit awareness: free-tier APIs have volume caps — EnergyPriceAPI (100 req/month), OilPriceAPI (unlimited) — monitor and alert
- No service reads from another service directly — all inter-service communication via PostgreSQL or VAULT API endpoint

5.2 JSON Output Schema — Draft (Dimitris to finalize)

```
{  "source_id": "EIA_WTI_DAILY",  "instrument": "WTI_CRUDE",  "delivery_point": "CUSHING_OK",  "timestamp": "2026-05-20T17:00:00Z",  "data_type": "SPOT_PRICE",  "value": 84.23,  "unit": "USD_PER_BBL",  "basis_vs_futures": -0.42,  "futures_contract": "CLM26",  "tos_status": "GO",  "gatekeeper_cleared": true,  "gatekeeper_id": "PARK_JAEHONG",  "raw_source_url": "api.eia.gov/..." }
```

Note: Schema above is a working draft. Dimitris has final authority on field names and structure. Park does not deploy microservices to production until Dimitris has reviewed and approved the schema.

6. Data Gatekeeper Role

The Gatekeeper role is separate from the Developer role. As Gatekeeper, Park is responsible for ensuring that no data source enters production Metrixx AI output without a documented TOS clearance decision. This is a compliance and data integrity function — not a technical function.

6.1 Gatekeeper Responsibilities

- Maintain catalog.json as the single source of truth for all Metrixx data sources
- Every source must have: tos_status (GO / HOLD / REVIEW / RED), last_reviewed date, review_notes, and gatekeeper_sign_off
- No new source may be added to any microservice without a catalog.json entry with Park's gatekeeper sign-off
- RED FLAG sources (Argus, full Platts API) are hard-blocked — raise to Jim immediately if any team member attempts to use them
- HOLD sources are blocked pending external resolution — track resolution status and update catalog.json when status changes
- Zillini is Publisher of Record: all AI-generated BASIS SENTIMENT outputs carry attribution data derived from Park's catalog

6.2 TOS Decision Matrix — Current Status

SOURCE	STATUS	CLEARED FOR	NOTES / ACTION
EIA API	GO	All Metrixx outputs	U.S. government open data. No restrictions. Build immediately.
USDA AMS API	GO	All Metrixx outputs	USDA open data. No restrictions. Build immediately.
CFTC data.cftc.gov	GO	All Metrixx outputs	CFTC open data. No restrictions. Weekly COT pull cleared.
FRED (St. Louis Fed)	GO	All Metrixx outputs	Federal Reserve open data. No restrictions.
Baker Hughes Rig Count	GO	All Metrixx outputs	Free public weekly publication. Paraphrase narrative, ingest data. No restrictions identified.
CME Public Settlements	GO	Internal analytics only	Daily settlement data is public. CME ILA applies to redistribution of raw tick data — settlements are not tick data. Park to confirm with Dimitris before using in any subscriber-facing output.
OilPriceAPI	GO	Demo / Alpha use	Free forever tier. TOS reviewed — no AI restriction identified. Confirm on registration.
Reuters RSS (free)	REVIEW	Event flagging / paraphrase only	Free RSS headline ingestion for event detection. No full article reproduction. Paraphrase only in any AI output. Legal review before production deployment.
Barchart cmdtyView	HOLD	BLOCKED pending agreement	Metrixx approved data partner. Sandbox pending. Do not use in any AI output until Barchart agreement signed by Jim.
S&P Global Platts	HOLD	BLOCKED pending licensing	No explicit AI prohibition found — but requires formal AI licensing confirmation before any use in Metrixx output. Processing Delegate model (Bloomberg users only) may provide an interim path.

Argus Media (full data)	RED	BLOCKED — DO NOT USE	EXPLICIT AI INGESTION PROHIBITION — January 2025 policy update. Argus free headline news (paraphrase only) cleared as T3. No Argus assessed prices under any circumstance without formal Argus AI license.
ICIS	REVIEW	BLOCKED pending legal review	No explicit AI restriction found. ICE Connect terms restrict derivative redistribution. Legal review required. Do not use in production output pending clearance.

7. Acceptance Criteria — June 1 Demo Ready

The following criteria define a successful WO-PARK-01 completion. All items must be satisfied before the June 12 Demo Day. Jim Marzano has final sign-off.

#	CRITERION	VERIFIED BY
1	EIA microservice deployed, running on cron, writing normalized JSON to PostgreSQL daily	Dimitris (architecture) + Jim (data accuracy)
2	USDA AMS microservice deployed, grain elevator bids for corn/soy/wheat in PostgreSQL by 8am CT daily	Mao (validates physical accuracy vs. market knowledge)
3	CFTC Disaggregated COT pull running every Friday, structured JSON by commodity delivered to Wei analytics layer	Wei (analytics) + Mao (COT interpretation)
4	Baker Hughes rig count weekly pull automated, -10 or more rig trigger flag operational	Jim (validates trigger logic)
5	Jet fuel Europe case study: end-to-end basis alert generated from EIA + ICE + Reuters data, producing valid 50-char / 50-word Claude output	Jim (commercial accuracy) + Mao (market validation)
6	catalog.json complete: all 30+ sources with TOS status, last reviewed date, and Park gatekeeper sign-off	Jim countersignature required
7	VAULT integration: completed basis calculations writing to VAULT with instrument tag, delivery point, date, and source provenance	Liu (DevOps deployment) + Jim (VAULT recall test)
8	No Tier 4 or Tier 5 source data present in any microservice output or Claude-generated subscriber output	Park (gatekeeper certification) + Jim (final audit)